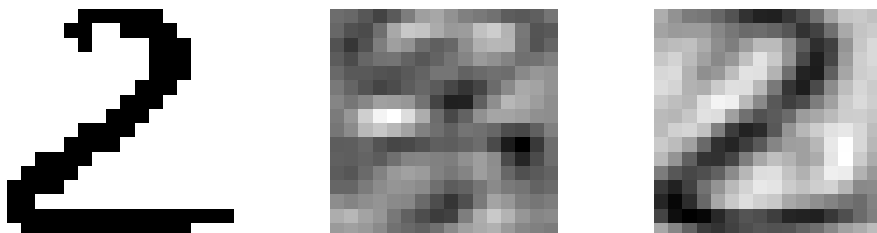


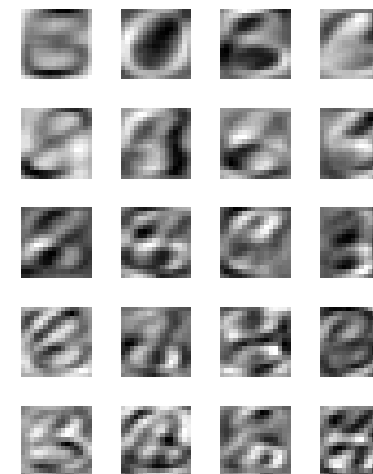
Principal Component Analysis (PCA)

1.6 -1.1 -1.6 2.1 -0.52 2.8 0.72 0.7 -0.68 -0.41 -1.4 -1.5 -0.54 -0.62 1.3 -1.4 -0.27 0.74 0.77 -1



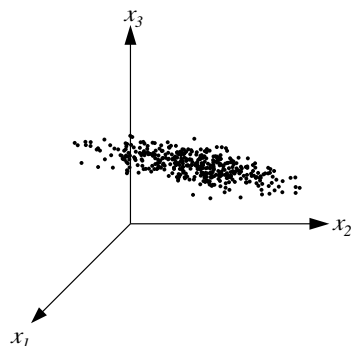
Covariance matrices, dimensionality reduction, PCA, Duality

1. **Covariance Matrices**
2. Principal Component Analysis
3. Duality



Spread of Data

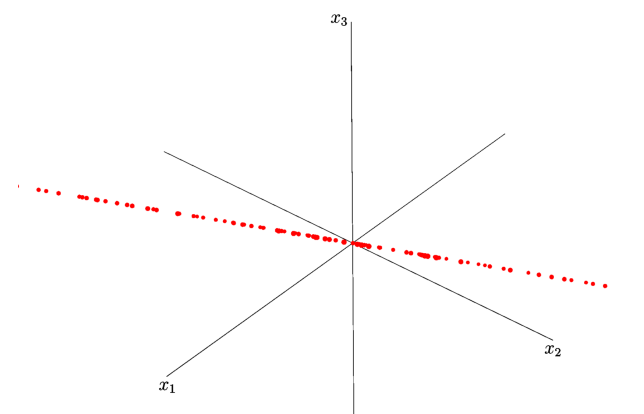
- Often data varies significantly in only some directions



- Reduce dimensions by projecting onto low dimensional subspace with maximum variation

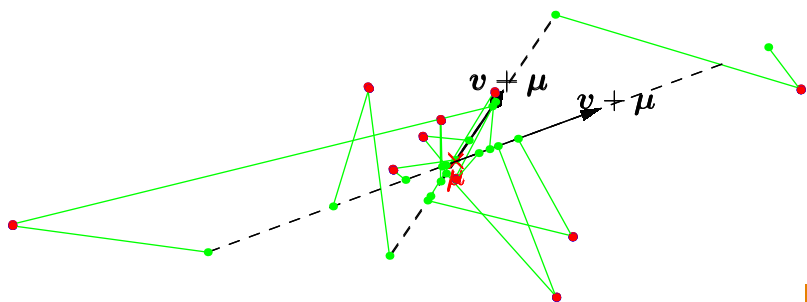
Looking is not Enough

Can't spot low dimensional data by looking at numbers



Dimensionality Reduction

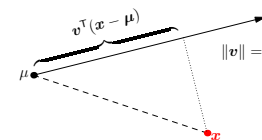
- Often helpful to consider only directions where data varies significantly
- Want to find directions along which data has its greatest variation



Direction of Maximum Variation

- Look for the vector v with $\|v\|^2 = 1$ to maximise

$$\sigma^2 = \frac{1}{m-1} \sum_{i=1}^m (v^\top (x_i - \mu))^2$$



- This is a constrained optimisation problem
- Solve by maximising Lagrangian

$$\mathcal{L} = \frac{1}{m-1} \sum_{k=1}^m (v^\top (x_k - \mu))^2 - \lambda (\|v\|^2 - 1)$$

- λ is a Lagrange multiplier

Direction of Maximum Variation

- Expanding the Lagrangian

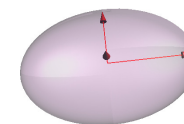
$$\begin{aligned} \mathcal{L} &= \frac{1}{m-1} \sum_{k=1}^m (v^\top (x_k - \mu))^2 - \lambda (\|v\|^2 - 1) \\ &= \frac{1}{m-1} \sum_{k=1}^m (v^\top (x_k - \mu) (x_k - \mu)^\top v) - \lambda (\|v\|^2 - 1) \\ &= v^\top \left(\frac{1}{m-1} \sum_{k=1}^m (x_k - \mu) (x_k - \mu)^\top \right) v - \lambda (\|v\|^2 - 1) \\ &= v^\top C v - \lambda (v^\top v - 1) \end{aligned}$$

- Extrema of the Lagrangian

$$\nabla \mathcal{L} = 2(Cv - \lambda v) = 0 \Rightarrow Cv = \lambda v$$

Direction of Maximum Variation

- The eigenvectors are directions that are extrema of the variance



- The variance in direction v is equal to

$$\begin{aligned} \sigma^2 &= \frac{1}{m-1} \sum_{i=1}^m (v^\top (x_i - \mu))^2 \\ &= v^\top C v = \lambda v^\top v = \lambda \end{aligned}$$

- The variance is maximised by the eigenvector with the maximum eigenvalue

Covariance Matrix

- The **covariance matrix** is defined as

$$\mathbf{C} = \frac{1}{m-1} \sum_{k=1}^m (\mathbf{x}_k - \boldsymbol{\mu})(\mathbf{x}_k - \boldsymbol{\mu})^\top$$

- The components C_{ij} measure how the i^{th} and j^{th} components co-vary

$$C_{ij} = \frac{1}{m-1} \sum_{k=1}^m (x_{ik} - \mu_i)(x_{jk} - \mu_j)$$

- C.f. covariance of random variables

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

Outer Product

- Remember that the outer-product of two vectors is defined as

$$\mathbf{x}\mathbf{y}^\top = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \begin{pmatrix} y_1 & y_2 & \cdots & y_n \end{pmatrix} = \begin{pmatrix} x_1 y_1 & x_1 y_2 & \cdots & x_1 y_n \\ x_2 y_1 & x_2 y_2 & \cdots & x_2 y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_n y_1 & x_n y_2 & \cdots & x_n y_n \end{pmatrix}$$

- C.f. Inner product

$$\mathbf{x}^\top \mathbf{y} = \begin{pmatrix} x_1 & x_2 & \cdots & x_n \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n$$

Matrix Form

- The covariance matrix is

$$\mathbf{C} = \frac{1}{m-1} \sum_{k=1}^m (\mathbf{x}_k - \boldsymbol{\mu})(\mathbf{x}_k - \boldsymbol{\mu})^\top$$

- Define the matrix

$$\mathbf{X} = \frac{1}{\sqrt{m-1}} (\mathbf{x}_1 - \boldsymbol{\mu}, \mathbf{x}_2 - \boldsymbol{\mu}, \dots, \mathbf{x}_m - \boldsymbol{\mu})$$

- We can write the covariance matrix as

$$\mathbf{C} = \mathbf{X}\mathbf{X}^\top$$

Properties of Covariance Matrix

- The **quadratic form** of a vector and matrix is defined as

$$\mathbf{v}^\top \mathbf{M} \mathbf{v}$$

- The quadratic form of a covariance matrix is non-negative for any vector

$$\mathbf{v}^\top \mathbf{C} \mathbf{v} = \mathbf{v}^\top \mathbf{X} \mathbf{X}^\top \mathbf{v} = \mathbf{u}^\top \mathbf{u} = \|\mathbf{u}\|^2 \geq 0$$

where $\mathbf{u} = \mathbf{X}^\top \mathbf{v}$

- Matrices with non-negative quadratic forms are known as **positive semi-definite**

Eigenvalue Decomposition

- The eigenvectors of $\mathbf{C}$ with the largest eigenvalues are known as the **principal components**
- The eigenvalues are all greater than or equal to zero
- Recall an eigenvector $\mathbf{v}$ satisfies the equation

$$\mathbf{C}\mathbf{v} = \lambda\mathbf{v}$$

- Multiplying both sides by $\mathbf{v}^T$

$$\mathbf{v}^T\mathbf{C}\mathbf{v} = \lambda\mathbf{v}^T\mathbf{v} = \lambda\|\mathbf{v}\|^2$$

but $\mathbf{v}^T\mathbf{C}\mathbf{v} \geq 0$ and $\|\mathbf{v}\|^2 > 0$ so

$$\lambda = \frac{\mathbf{v}^T\mathbf{C}\mathbf{v}}{\|\mathbf{v}\|^2} \geq 0$$

Surface Defined by Matrix

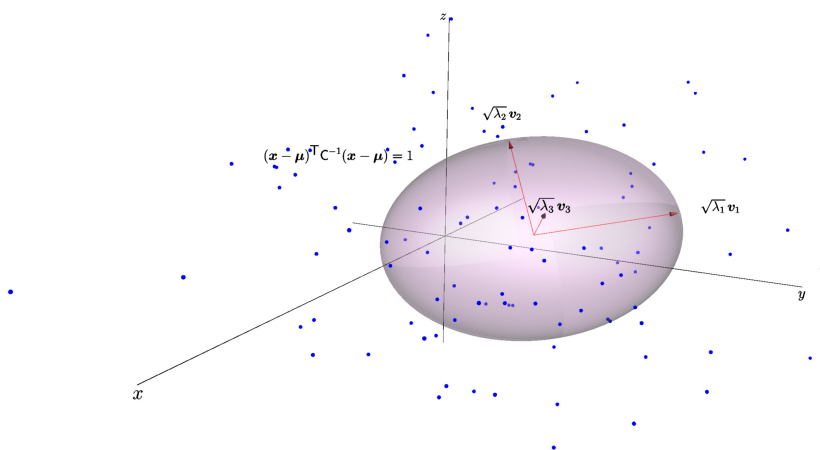
- The set of vectors $\mathbf{x}$ such that

$$\mathbf{x}^T\mathbf{C}^{-1}\mathbf{x} = 1$$

defines a surface

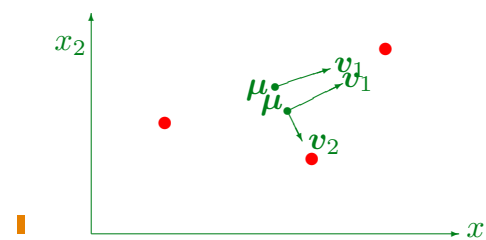
- The surface is an ellipsoid, $\mathcal{E}$
- The eigenvectors point in the direction of the principal axes of the ellipsoid
- The radii of the principal axes are equal to the square root of the eigenvalues

Ellipsoid and Eigen Space



Spanning Input Space

- A covariance matrix will have a zero eigenvalue only if there is no variation in the direction of the corresponding eigenvector
- A covariance matrix will have zero eigenvalues if the number of patterns are less than or equal to the number of dimensions
- A covariance matrix formed from $p + 1$ patterns that are linearly independent (i.e. you cannot form any one out of p of the other patterns) will have no zero eigenvalues

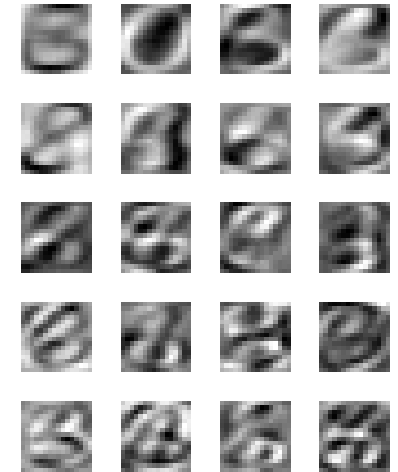


Positive Definite

- Matrices with no zero eigenvalues are called **full rank** matrices (as opposed to rank deficient)
- Full rank matrices are invertible, rank deficient matrices are singular and non-invertible
- Full rank covariance matrices that have positive eigenvalues only and are said to be **positive definite**
- We would expect that when $m > p$ the covariance matrix will be positive definite unless there are some symmetries that linearly constrain the patterns

Outline

1. Covariance Matrices
2. **Principal Component Analysis**
3. Duality



Principal Component Analysis

- PCA occurs as follows
 - ★ Construct the covariance matrix
 - ★ Find the eigenvalues and eigenvectors
 - ★ Keep the eigenvectors with the largest eigenvalues (principal components)
 - ★ Project the inputs into the space spanned by the principal components
- We then use the projected inputs as inputs to our learning machine

Projection Matrix

- To project the inputs construct the projection matrix

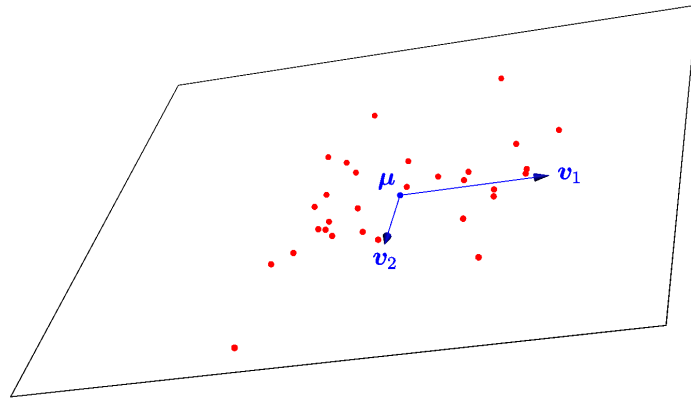
$$\mathbf{P} = \begin{pmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_k^T \end{pmatrix}$$

- $k < p$ is the number of principal components we keep
- Given a p -dimensional input pattern $\mathbf{x}$ we can construct a k -dimensional representation $\mathbf{z}$

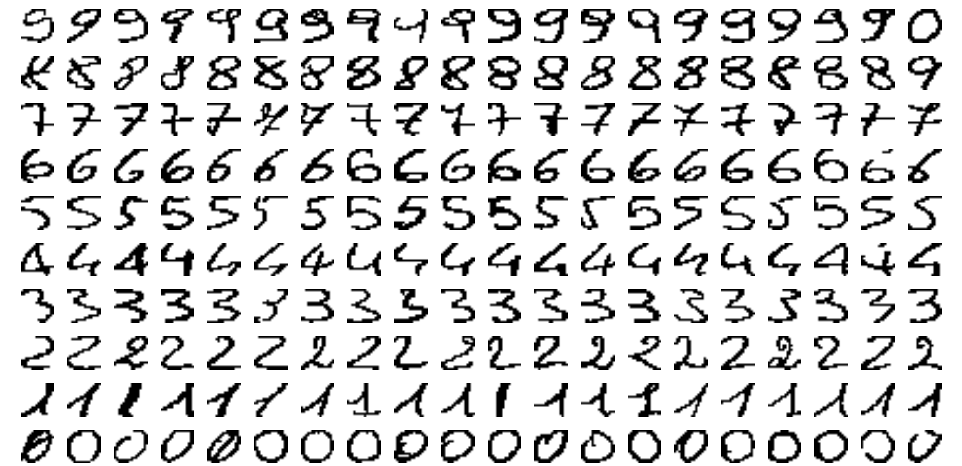
$$\mathbf{z} = \mathbf{P}(\mathbf{x} - \boldsymbol{\mu})$$

- Use $\mathbf{z}$ as our new inputs

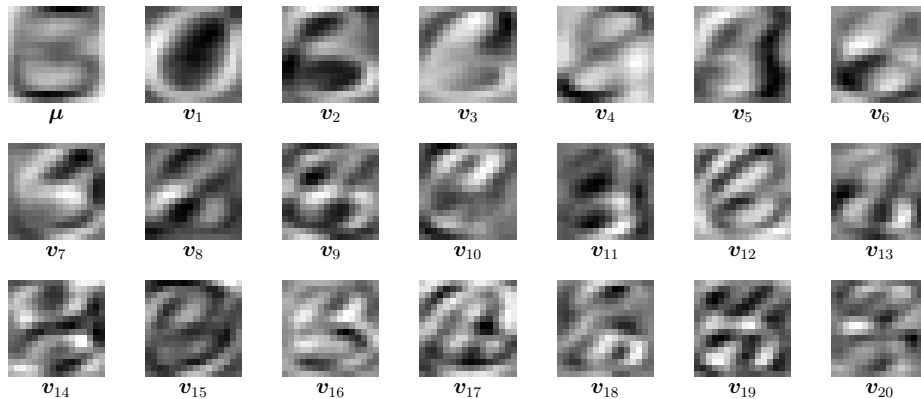
Subspace Projection



Hand Written Digits



Eigenvectors



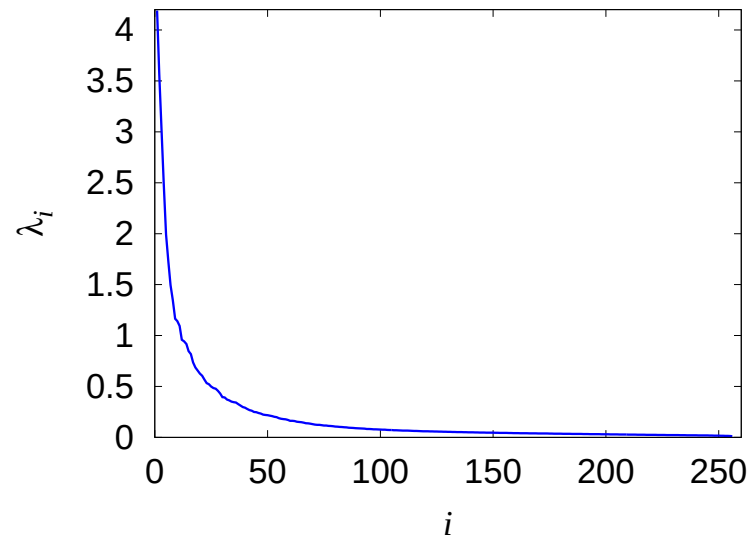
Reconstruction

- Projecting into a subspace of eigenvectors can be seen as approximating the inputs by

$$\hat{x}_i = \mu + \sum_{j=1}^k z_j^i v_j, \quad z_j^i = v_j^T (x_i - \mu), \quad \|v_j\| = 1$$

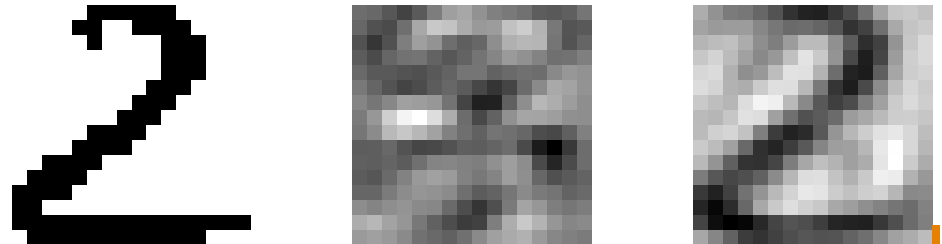
- Principle component analysis projects the data into a subspace of size m with the minimal approximation error $\mathbb{E}[\|\hat{x}_i - x_i\|^2]$
- The loss of “energy” (or squared error) is equal to the sum of the eigenvalues in the directions that are ignored

Eigenvalues for Digits



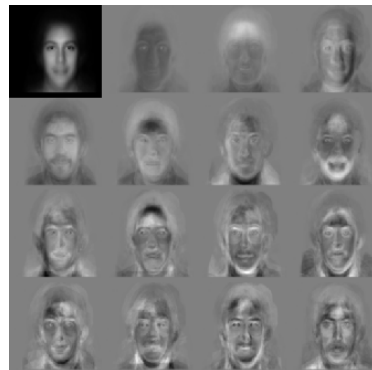
Reconstruction from Eigenvectors

1.6 -1.1 -1.6 2.1 -0.52 2.8 0.72 0.7 -0.68 -0.41 -1.4 -1.5 -0.54 -0.62 1.3 -1.4 -0.27 0.74 0.77 -1



Outline

1. Covariance Matrices
2. Principal Component Analysis
3. **Duality**



PCA for Images

- An image often contains around $p = 256 \times 256 = 64k$ pixels
- In standard PCA we would create an $p \times p$ matrix with over 4×10^9 elements
- This is intractable
- m images span at most a $m - 1$ dimensional subspace
- Usually this subspace will be much smaller than the space of all images $m \ll p$

Dual Matrix

- The covariance $\mathbf{C} = \mathbf{X}\mathbf{X}^\top$ is a $p \times p$ matrix
- Consider the $m \times m$ matrix $\mathbf{D} = \mathbf{X}^\top\mathbf{X}$
- Suppose $\mathbf{v}$ is an eigenvector of $\mathbf{D}$

$$\mathbf{D}\mathbf{v} = \lambda\mathbf{v}$$

$$\mathbf{X}^\top\mathbf{X}\mathbf{v} = \lambda\mathbf{v}$$

$$\mathbf{X}\mathbf{X}^\top\mathbf{X}\mathbf{v} = \lambda\mathbf{X}\mathbf{v}$$

$$\mathbf{C}\mathbf{X}\mathbf{v} = \lambda\mathbf{X}\mathbf{v} \Rightarrow \mathbf{C}\mathbf{u} = \lambda\mathbf{u}$$

- $\mathbf{u} = \mathbf{X}\mathbf{v}$ (and $\mathbf{v} \propto \mathbf{X}^\top\mathbf{u}$)

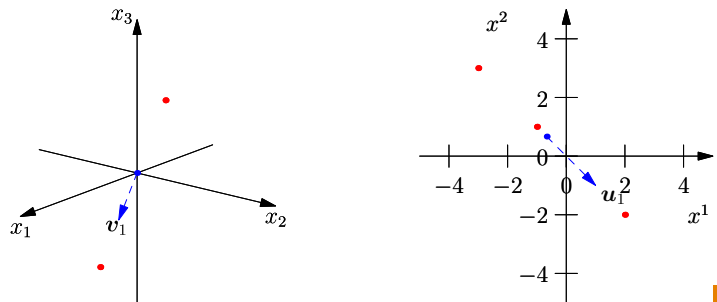
Dual Matrix

- Matrices $\mathbf{C} = \mathbf{X}\mathbf{X}^\top$ and $\mathbf{D} = \mathbf{X}^\top\mathbf{X}$ have the same eigenvalues
- Can use the dual $m \times m$ matrix $\mathbf{D}$ to find eigenvalues and eigenvectors of $\mathbf{C}$
- Note that $\mathbf{D} = \mathbf{X}^\top\mathbf{X}$ has components $D_{kl} \propto (\mathbf{x}_k - \boldsymbol{\mu})^\top(\mathbf{x}_l - \boldsymbol{\mu})$
- Takes $O(p \times m \times m)$ time to construct $\mathbf{D}$
- We work in a “dual space” which is the space spanned by the examples

What Does a Subspace Look Like?

- Consider $\mathbf{y}^1 = \begin{pmatrix} 2 \\ 4 \\ 4 \end{pmatrix}$, $\mathbf{y}^2 = \begin{pmatrix} 8 \\ 6 \\ 2 \end{pmatrix}$ with mean $\boldsymbol{\mu} = \begin{pmatrix} 5 \\ 5 \\ 3 \end{pmatrix}$
- Subtracting the mean $\mathbf{x}^i = \mathbf{y}^i - \boldsymbol{\mu}$ we can construct matrix

$$\mathbf{X} = \begin{pmatrix} x_1^1 & x_1^2 \\ x_2^1 & x_2^2 \\ x_3^1 & x_3^2 \end{pmatrix} = \begin{pmatrix} -3 & 3 \\ -1 & 1 \\ 2 & -2 \end{pmatrix}$$



Summary

- PCA allows us to reduce the dimensionality of the inputs
- We project the inputs into a sub-space where the data varies the most
- We can work in either the original space ($\mathbf{X}\mathbf{X}^\top$) or the dual space ($\mathbf{X}^\top\mathbf{X}$)
- When we have many more features than examples (i.e. $p \gg m$) then it is more efficient working in the dual space
- We will see examples of dual spaces again when we look at SVMs